
Observability Debt in Agent Evaluation: What a Telemetry Layer for Agentic OS Must Capture

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Abstract

1 Agentic AI systems are evaluated today through three disjoint public infrastructures—
2 leaderboards, trace corpora, and telemetry logs—none of which jointly exposes the fields
3 a resource-aware operating layer would need: task identity, model identity, scaffold iden-
4 tity, outcome, token/cache usage, retries, latency, and billing. We treat this fragmenta-
5 tion as an empirical object and measure its consequences across three studies on public
6 agent-evaluation data. First, using the Holistic Agent Leaderboard (HAL), raw dollar-cost
7 rankings transfer across some benchmark pairs (Spearman $\rho=0.74\text{--}0.93$) but are largely
8 explained by a stable label-level cost propensity (pooled model $R^2=82.0\%$); success-
9 adjusted cost is markedly less portable (mean $\rho=0.59$ vs. 0.84). Second, using 977 exact
10 task–model–matched trajectory pairs from Open-SWE-Traces, scaffold choice changes ob-
11 servable execution burden at fixed task and model (OpenHands/SWE-agent turn ratio 0.85 ,
12 $[0.82, 0.87]$), and the widely assumed “expensive failure” premium is not detected under
13 exact task matching (80%-power threshold: premium fractions ≥ 0.65 ; observed 0.47--
14 0.53). Third, auditing two independent telemetry corpora (8,058 and 1,017 sessions), we
15 characterize the resource anatomy—93.7% mean cache-read share, $4.29\times$ conditional tool-
16 error cascades replicated across sources, top-1%-of-sessions holding 48.3% of input-token
17 volume—that a scheduler or router would need but that no benchmark currently attaches to
18 an outcome. Together, these studies show that every unconditioned aggregate computable
19 from today’s public data conflates *which workloads were attempted with how agents be-*
20 *haved on them*. We specify the minimal joint telemetry schema this evidence implies and
21 argue it belongs at the operating-system layer.

22 1 Why This Belongs at the OS Layer

23 Agentic AI systems persist state, invoke tools, retry failed actions, and accumulate context across long horizons.
24 Reasoning about their resource behavior—for scheduling, routing, cost budgeting, or governance—requires the
25 same unified telemetry that operating systems have long provided for conventional workloads: a common
26 schema linking *what ran*, *on what*, *under which policy*, *with what outcome*, and *at what resource cost*. No such
27 schema currently exists for agentic AI, and we show empirically that its absence is the dominant source of noise
28 in every cost signal practitioners rely on.

29 We treat three public data sources as stand-ins for what an agentic OS observability layer would unify: a cost-
30 and-outcome leaderboard (HAL), a task/model/scaffold-matched trajectory corpus (Open-SWE-Traces), and
31 raw session-level usage telemetry (two independent corpora). Each exposes a different partial view. HAL links
32 dollar cost to outcome but encrypts execution traces. Open-SWE-Traces exposes exact task/model/scaffold
33 matching and resolved status but no tokens or billing. The telemetry corpora expose token, cache, and tool-
34 call dynamics but no outcome or billing. **No public source jointly exposes task, model, scaffold, outcome,**
35 **token/cache usage, retries, latency, and billing**—the minimal schema a resource-aware OS layer needs.

36 We make this concrete with three linked results: (1) a coverage-aware cost-portability study showing raw-cost
37 rankings conflate label-level pricing with benchmark difficulty while success-adjusted cost decorrelates; (2) a
38 matched-trajectory study showing scaffold burden differences are real but the “failures cost more” heuristic

measures workload composition; and (3) a cross-source telemetry audit characterizing the context growth, cache dominance, and error cascades a scheduler would need to observe directly—none of which can currently be attached to an outcome or a dollar figure.

2 Related Work

Kapoor et al. (2026) introduce HAL as cost-aware evaluation infrastructure spanning coding, web, science, and customer-service tasks (21,730 rollouts, roughly \$40,000 in evaluation cost). Kapoor et al. (2024) motivate joint accuracy–cost reporting and convex-hull frontier analysis, which we adopt as one of three decision-frontier definitions. Ndzomga (2026) asks whether task subsets preserve agent rankings within a single benchmark; we instead ask whether an *observed cost signal* predicts a *different* workload. Benchmark² studies cross-benchmark rank consistency for static LLM benchmarks; our setting adds interactive scaffolds, dollar cost, success adjustment, and a held-out workload test. SWE-Effi reports that failed attempts consume substantially more resources than successful ones; our matched design tests that claim at fixed task and does not detect the premium once task-difficulty confounding is removed (Section 4). Open-SWE-Traces supplies the exact matching that makes this test possible but lacks token/billing fields, motivating Section 6. Our contribution is a demonstration—not an assumption—that fragmented telemetry actively distorts the signals practitioners compute, plus a schema specification grounded in that measurement.

3 Study 1: Does a Cost Signal Travel Across Workloads?

3.1 Data and cohort

We use the public `all_leaderboards_costs_HAL.csv` snapshot distributed with Ndzomga (2026): 242 displayed evaluations across nine benchmarks, 11 scaffolds, and 29 displayed model labels. The CSV supplies accuracy and aggregate dollar cost but no token components, provider aliases, tool-call counts, retries, or per-task trajectories—the exact gap this paper is about. Most evaluations are single-run, so rollout-level uncertainty is unavailable.

The only scaffold repeated broadly enough is the displayed HAL Generalist Agent. Fixing it and matching exact label strings yields six workloads with adequate overlap—CORE-Bench Hard, GAIA, SciCode, ScienceAgentBench, SWE-bench Verified Mini, TAU-bench Airline—for a cohort of 86 rows and 25 labels (USACO has one Generalist observation and is excluded).

3.2 Raw cost transfers; the mechanism behind it does not

Across six pairs with ≥ 12 shared labels, raw dollar-cost Spearman ρ ranges 0.74–0.93 (mean 0.84), surviving Holm-corrected permutation testing (Figure 1). A pooled label-plus-benchmark log-cost fixed-effects model explains $R^2=82.0\%$ (adjusted 72.7%); once label and benchmark propensity are removed, **no high-overlap residual-cost association survives** a 20,000-draw permutation test.

Leave-one-benchmark-out (LOBO) prediction succeeds for CORE ($\rho=0.78$, $N=14$, $p=0.0016$), GAIA (0.87, $N=16$), SWE-mini (0.90, $N=16$), TAU (0.83, $N=14$), all robust to every leave-one-label-out omission (e.g., CORE [0.73, 0.91], SWE-mini [0.88, 0.92]). It fails for SciCode (unstable, $[-0.50, 0.14]$; 2.1% mean accuracy, one-third zero-accuracy labels) and ScienceAgentBench (directionally negative, $[-0.77, -0.43]$; narrowest log-cost dispersion, $N=7$).

Interpretation: the portable component is a stable label-level propensity—plausibly provider pricing, reasoning-mode defaults, or routing policy—that a router could learn and reuse. But public data cannot identify *which* mechanism it is, because none of the distinguishing fields (token components, routing decisions, cache treatment) are exposed.

3.3 Success-adjusted cost is a different, less portable ranking

Cost per expected success, $\text{CPS}_{i,b}(\epsilon) = C_{i,b} / \max(A_{i,b}, \epsilon)$ with primary floor $\epsilon=0.01$ (sensitivities at 0.001, 0.05, and zero-exclusion in the supplement), has mean $\rho=0.59$. Only three pairs survive correction: CORE–GAIA (0.87, [0.63, 0.96]), CORE–TAU (0.83, [0.31, 1.00]), GAIA–TAU (0.78, [0.42, 0.94]). The clearest raw-vs-CPS gap is GAIA–SWE-mini (0.60, [0.14, 1.13]): a label cheap in dollars is not reliably cheap per expected success elsewhere. No label tested on ≥ 5 workloads sits on every accuracy–cost frontier under discrete nondominance, convex hull, or 5% tolerance (o4-mini Low reaches 5/6 under tolerance; no universal winner under any strict definition).

Implication: routing on displayed dollar cost optimizes a signal that decorrelates from efficiency exactly where procurement decisions matter.

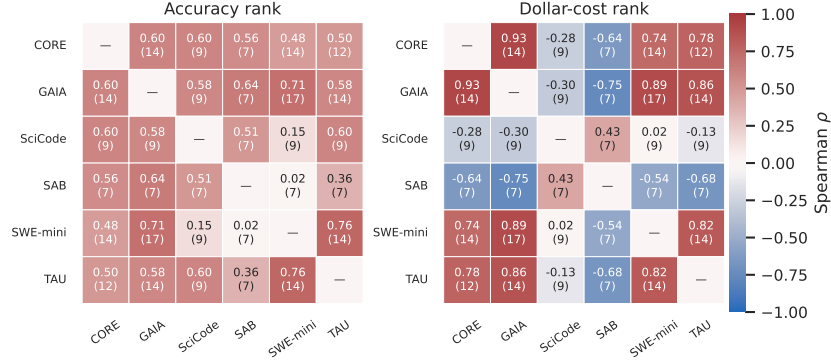


Figure 1: **Raw cost transfers; success-adjusted cost does not.** Pairwise Spearman rank correlations across benchmark pairs sharing ≥ 12 displayed labels. Raw dollar cost is strongly portable (mean 0.84); cost per expected success is not (mean 0.59), and only three CPS pairs survive Holm-corrected permutation testing.

4 Study 2: What Matched Trajectories Reveal That Aggregates Cannot

HAL’s own trace archives are encrypted, so Study 1 cannot say *why* costs differ. We use an external unencrypted corpus—977 exact task–model pairs from Open-SWE-Traces for MiniMax-M2.5 under two scaffolds (OpenHands, SWE-agent), a documented four-shard boundary sample that is reproducible but not random. The schema exposes resolved status, roles, tool calls, and text but not billed tokens or dollars; we treat turns, tool calls, and characters as trace-burden proxies.

4.1 Scaffold choice changes observable burden at fixed task

OpenHands has lower matched medians: 123 vs. 149 trajectory turns (ratio 0.85, $[0.82, 0.87]$), 61 vs. 74 tool calls (0.85, $[0.82, 0.86]$), 176,708 vs. 194,263 characters (0.95, $[0.92, 0.98]$); it is lower on 69.7% of pairs (Figure 2a). Resolved status shows no detected difference ($p=0.197$): scaffolds differ in execution structure without a detected accuracy trade-off here.

4.2 The “expensive failure” premium is a composition artifact

Comparing failed vs. resolved trajectories *across* tasks reproduces SWE-Effi’s pattern (19–36% heavier failures). But among the **107 discordant pairs**—identical task, identical model, one scaffold resolved and the other failed—the failed side is heavier only 47–52 times out of 107 on every metric (two-sided binomial $p \geq 0.25$; observed fraction CIs $[0.37, 0.57]$ raw, $[0.43, 0.63]$ normalized). Normalizing each trajectory by its scaffold’s median burden gives failed/resolved ratios of 0.98–1.06 spanning 1. A pre-specified power calculation gives 80% power against a premium fraction ≥ 0.65 (failed heavier on 70+ of 107); observed 0.47–0.53 rules out strong versions but not modest ones ($< 15pp$). One model, nonrandom sample: this tests the heuristic here, not its general absence.

Interpretation: hard tasks make agents both burn more *and* fail more. An unconditioned failure multiplier measures *which tasks were attempted*, not a property of failing runs. Task-conditioned accounting requires exactly the joint schema no single public source provides.

5 Study 3: The Telemetry an OS Layer Would Need to See Directly

Setting aside outcome and billing entirely: what does raw execution telemetry look like? We audit two independent coding-agent corpora (Table 1). **TraceLab:** 8,058 real-world sessions, 665,453 LLM rounds, 743,819 tool calls. **MIMO Claude Code:** 1,017 sessions, 4,690 rounds, single model. Neither links usage to outcome or billing; all summaries are compositional.

Context is the dominant resource, not generation. The median TraceLab round carries 132,092 input tokens against 249 output tokens (93.7% mean cache-read share); median final-round context is $3.6\times$ initial (P90 $9.9\times$). A leaderboard dollar total cannot distinguish price-driven from cache-policy- or accumulation-driven spend. Tool verbosity is one mechanism: when a round carries $< 1k$ tool-result characters, median next-input growth is 443 tokens; at 1k–10k it is 1,412; above 10k it is 5,420.

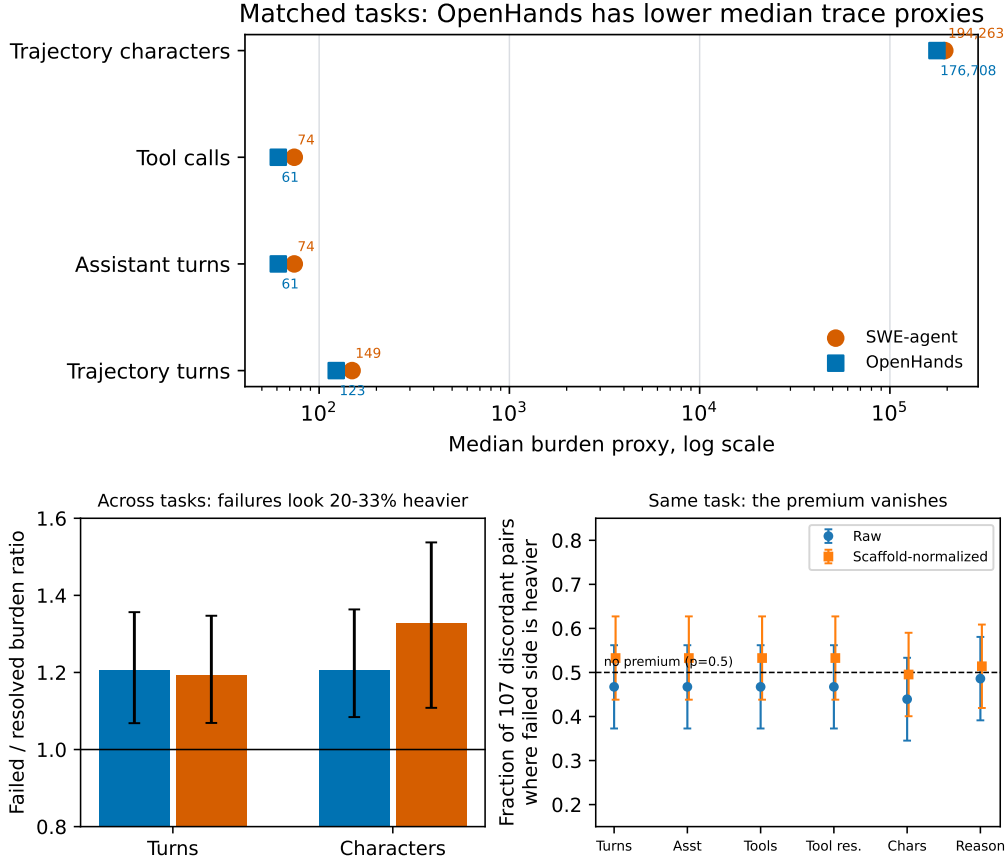


Figure 2: (a) Scaffold burden on 977 exact task-model pairs: OpenHands uses $\sim 15\%$ fewer median turns/tool calls than SWE-agent at fixed task and model. (b) The failure premium dissolves under exact matching. Left: naive cross-task failed/resolved ratios reproduce folklore (19–36% heavier failures). Right: among 107 discordant pairs (same task, opposite outcomes), the failed side is heavier only 47–52/107 on every metric.

123 **Resource burden is heavy-tailed.** The top 1% of sessions hold 48.3% of summed input volume; top 10%
 124 hold 88.2% (Figure 3). Median session input rises from 0.34M tokens at zero observed tool errors to 14.65M at
 125 six or more. A scheduler budgeting against mean cost would be badly calibrated for the sessions that dominate
 126 spend.

127 **Error cascades replicate across independent sources.** After a successful tool call, the next call errs 4.9%;
 128 after an error, 21.0% ($4.29\times$). MIMO—different model, different site—replicates at $4.33\times$ (4.3% vs. 1.0%).
 129 Retrying is the norm (88.3% of errored calls get a same-tool retry; 77.9% succeed), but the tail is pathological:
 130 consecutive-error runs have median length 1, P90 3, max 368. Failures are also slower yet less informative:
 131 median latency 313 ms vs. 140 ms (P95 85.6 s vs. 15.8 s) while returning 208 vs. 602 characters—more wall
 132 time, less signal.

133 **Data-quality boundary.** Every released MIMO session repeats one constant per-round usage tuple, so within-
 134 session token dynamics cannot be validated there; we restrict MIMO to session counts, corpus totals, cross-
 135 session concentration, and content-derived error sequences.

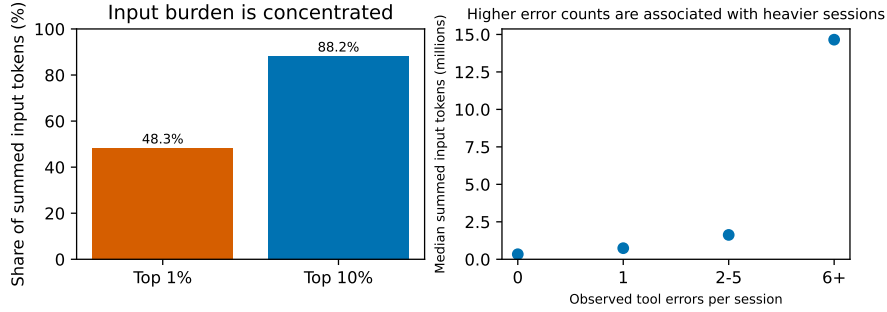
136 6 The Minimal Joint Schema

137 Table 2 summarizes the coverage gap directly. No column is complete; the missing rows are precisely those
 138 Sections 3–5 show to be load-bearing. Based on these gaps, the minimal joint schema per execution unit is:

- 139 1. **Task and model identity** (exact IDs, not display strings)—enables exact-matched designs; without it the
 140 Section 4 test is impossible.

Table 1: Telemetry anatomy across two independent corpora.

	TraceLab	MIMO Claude Code
Sessions / LLM rounds	8,058 / 665,453	1,017 / 4,690
Median input / output tokens per round	132,092 / 249	n/a (constant-tuple audit)
Mean cache-read share	93.7%	cache-read = $16.3\times$ uncached
Median final-vs-initial context	$3.6\times$ (P90 $9.9\times$)	n/a
Top-1% / top-10% input-token share	48.3% / 88.2%	— / 37.9%
Error cascade $P(\text{err} \text{err})/P(\text{err} \text{succ})$	$21.0/4.9\% = 4.29\times$	$4.3/1.0\% = 4.33\times$
Errored calls retried same-tool / retry succeeds	88.3% / 77.9%	—
Consecutive-error run length (median/P90/max)	1 / 3 / 368	—
Tool-call latency, success vs. error (median)	140 ms / 313 ms	—
Tool-result size, success vs. error (median chars)	602 / 208	—

Figure 3: **Heavy-tailed burden.** Top-1%/top-10% concentration of summed input tokens (left axis) and median session input by observed-error bin (right): 0.34M at zero errors rising to 14.65M at 6+.

- 141 2. **Scaffold/harness identity and version**—attributes burden differences to scaffold design rather than model
142 behavior; input to routing.
- 143 3. **Outcome/resolved status**—conditions any cost statistic on success, closing the CPS portability gap.
- 144 4. **Token components per round (input, output, cache-read, cache-creation)**—separates price from accu-
145 mulation and cache policy; lets a memory layer budget context growth.
- 146 5. **Tool-call counts, retries, error status per call**—detects and prices $4.29\times$ cascade dynamics; input to
147 governance deciding when to halt a spiraling session.
- 148 6. **Latency per call and per session**—present only as unlinked tool-call durations in one corpus; required for
149 real-time scheduling.
- 150 7. **Billing, linked to all of the above**—the one field HAL exposes well, and exactly the field that is uninter-
151 pretable without the others.

152 No existing release exposes this joint schema. We do not propose a storage format or API—a systems-
153 design question outside our scope—but the measured consequences of its absence argue it belongs at a shared
154 operating-system layer rather than being re-derived piecemeal inside each benchmark, where today it is not
155 derived at all.

156 7 Limits

157 This is a secondary-data analysis with clear construct boundaries. Display labels do not fully identify prompts,
158 tools, budgets, harness versions, or routing; the HAL cohort is sparse, unbalanced, mostly single-run. Propensity
159 adjustment is observational decomposition, not list-price control; CPS is an aggregate ratio, not the mean cost
160 of successful runs. Study 2 uses one model and a nonrandom four-shard sample whose proxies are not tokens
161 or dollars; the null is powered only against strong premiums ($\geq 15\text{pp}$). Study 3 is observational, unlinked
162 to outcomes or billing; MIMO’s per-round fields are unverifiable and used only where supported. None of
163 this invalidates the schema argument—each limit is itself an instance of the missing-field problem the paper
164 documents.

Table 2: **The observability gap.** Field coverage per public source: ■ exposed, □ partial (display-label-only, degraded fidelity, or inferable), – absent. No source covers the joint schema; the union still leaves latency unlinked and round-level token fidelity unverifiable outside one corpus.

Field	HAL	Open-SWE	TraceLab	MIMO
Task identity	□	■	–	–
Model identity	□	■	□	■
Scaffold identity/version	■	■	□	■
Outcome / resolved status	■	■	–	–
Tokens / cache components	–	–	■	□
Retries / error status	–	□	■	□
Latency	–	–	■	–
Billing	■	–	–	–

8 Conclusion

Three independent studies converge: raw cost, success-adjusted cost, and failure premiums are all measurably confounded by information today’s infrastructure does not expose jointly. Raw cost transfers but mostly reflects stable label pricing; success adjustment reorders rankings and survives correction for half the tested pairs; the failure premium disappears under exact matching. Beneath all three sits one missing artifact: no public source jointly links task, model, scaffold, outcome, token/cache usage, retries, latency, and billing. We specified that schema and argued it belongs to a shared operating-system layer for agentic AI—the role resource accounting plays in conventional OSes—not something each leaderboard should reconstruct independently, incompletely, and incompatibly.

Checklist

- 1. Claims:** All claims are supported by analyses of public data with pre-specified tests, permutation/Holm correction where applicable, and explicit power statements (Section 4).
- 2. Limitations:** Section 7 discusses construct boundaries, sampling, proxy validity, and power.
- 3. Data:** All sources are public: HAL CSV (Ndzomga 2026), Open-SWE-Traces (Ahmad et al. 2026), TraceLab and MIMO telemetry corpora.
- 4. Code:** Analysis pipeline and frozen outputs will be released upon publication; results are reproducible from public data alone.
- 5. Ethics/anonymity:** Double-blind; secondary analysis of public evaluation artifacts; no human subjects; no identifying information.

References

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